

1 Mathematical Preliminaries

Reading guide. For the first lecture, focus primarily on Section 1.3: probability spaces, random variables, distributions, densities, expectation, and conditional density.

Sections 1.1–1.2 provide optional measure-theoretic background and are included for completeness. Markov chains and stochastic processes will become important in the diffusion-model lectures.

1.1 σ -algebras and measures

Definition 1 (σ -algebra). Let Ω be a set and $\mathcal{F}$ be a set of subsets of Ω , i.e. each of the element of $\mathcal{F}$ is a subset of Ω . Then $\mathcal{F}$ is called a **σ -algebra** (with the main set Ω) if the following conditions hold:

- ($\sigma 0$) The big set Ω is in $\mathcal{F}$, i.e. $\Omega \in \mathcal{F}$;
- ($\sigma 1$) $A \in \mathcal{F}$ follows that the complement $A^c = \Omega \setminus A = \{\omega \in \Omega \mid \omega \notin A\}$ is also in $\mathcal{F}$, i.e. $A^c \in \mathcal{F}$;
- ($\sigma 2$) for a (countable) sequence $A_1, A_2, \dots \in \mathcal{F}$ their union is also in $\mathcal{F}$:

$$\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}.$$

The pair $(\Omega, \mathcal{F})$ is called a **measurable space**.

Definition 2 (measurable function). Let $(\Omega_1, \mathcal{F}_1)$ and $(\Omega_2, \mathcal{F}_2)$ be measurable spaces, then a function

$$f : \Omega_1 \rightarrow \Omega_2$$

is called a **measurable function** if for any $B \in \mathcal{F}_2$ the set $f^{-1}(B) = \{\omega \in \Omega_1 \mid f(\omega) \in B\}$ is from $\mathcal{F}_1$.

Definition 3 (σ -algebras generated by a family of sets). Let Ω be a set, and $\mathcal{A}$ be a set of subsets of Ω . Then the **minimal**¹ σ -algebra containing $\mathcal{A}$ is called the **σ -algebra generated by $\mathcal{A}$** and is denoted by $\sigma(\mathcal{A})$. It can be shown that

$$\sigma(\mathcal{A}) = \bigcap_{\{\sigma\text{-algebra } \mathcal{F} \mid \mathcal{A} \subseteq \mathcal{F}\}} \mathcal{F}.$$

Definition 4 (Borel algebra, Borel sets). For $\Omega = \mathbb{R}^n$ the σ -algebra generated by all **open rectangles** is called **Borel algebra** and is denoted by $\mathcal{B}(\mathbb{R}^n)$:

$$\mathcal{B}(\mathbb{R}^n) = \sigma(\{(a_1, b_1) \times (a_2, b_2) \times \dots \times (a_n, b_n) \mid a_i, b_i \in \mathbb{R}, a_i < b_i\}).$$

Definition 5 (measure). Let $(\Omega, \mathcal{F})$ be a measurable space, then a function $\mu : \mathcal{F} \rightarrow \mathbb{R} \cup \{\infty\}$ is called a **measure** if

- (m1) for any set $E \in \mathcal{F}$ we have $\mu(E) \geq 0$;
- (m2) $\mu(\emptyset) = 0$;
- (m3) for any sets $E_1, E_2, \dots \in \mathcal{F}$ such that $E_i \cap E_j = \emptyset$ ($i \neq j$) we have

$$\mu\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \mu(E_i).$$

Then the triplet $(\Omega, \mathcal{F}, \mu)$ is called a **measure space**.

¹here a σ -algebra $\sigma(\mathcal{A})$ is said to be “minimal” if any σ -algebra $\mathcal{F}$ such that $\mathcal{A} \subseteq \mathcal{F}$, contains $\sigma(\mathcal{A})$: $\sigma(\mathcal{A}) \subseteq \mathcal{F}$

EXAMPLE (Dirac measure): Let $(\Omega, \mathcal{F})$ be a measurable space, $a \in \Omega$ be an element. Then the function

$$\delta_a : \mathcal{F} \rightarrow \mathbb{R} \cup \{\infty\}, \quad \delta_a(A) = \mathbb{1}_A(a) = \begin{cases} 1 & \text{if } a \in A \\ 0 & \text{otherwise} \end{cases}$$

is called **Dirac measure**, where $\mathbb{1}_A(\cdot)$ is the **indicator function** of the set A .

Exercise 1. Make sure that δ_a is a measure on the measurable space $(\Omega, \mathcal{F})$.

EXAMPLE (Lebesgue measure): Let $\Omega = \mathbb{R}$ (can be defined for $\mathbb{R}^n$, for any $n \in \mathbb{N}$), $I = (a, b)$, $a < b$, define for any such I interval the quantity $l(I) = b - a$ which will be the **length** of the interval I . Then for any subset $E \subseteq \mathbb{R}$ define the following quantities:

$$\lambda^*(E) = \inf \left\{ \sum_{k=1}^{\infty} l(I_k) \mid I_k = (a_k, b_k) \text{ such that } E \subseteq \bigcup_{k=1}^{\infty} I_k \right\},$$

$$\lambda_*(E) = \sup \left\{ \sum_{k=1}^{\infty} l(I_k) \mid I_k = (a_k, b_k) \text{ such that } E \supseteq \bigcup_{k=1}^{\infty} I_k \right\},$$

which are called **outer** and **inner** measures of E .

Definition 6 (Lebesgue measurable set). The set $E \subseteq \mathbb{R}$ is called **Lebesgue measurable** if $\lambda^*(E) = \lambda_*(E)$.

Exercise 2. Show that the set of all Lebesgue measurable sets is a σ -algebra. Why this σ -algebra contains $\mathcal{B}(\mathbb{R})$?

Definition 7 (Lebesgue measure). The σ -algebra of all Lebesgue-measurable sets is called **Lebesgue algebra**, and the measure $\lambda(E) = \lambda^*(E) = \lambda_*(E)$ is called **Lebesgue measure**.

1.2 Integrals

Let $(\Omega, \mathcal{F}, \mu)$ be a measure space, $S \subseteq \Omega$, and $\mathbb{1}_S$ be its indicator function, i.e. $\mathbb{1}_S(x) = 1$ if $x \in S$, and $\mathbb{1}_S(x) = 0$ if $x \notin S$.

Definition 8 (simple function). A function $f : \Omega \rightarrow \mathbb{R}$ is called a **simple function** if there are disjoint sets $S_1, S_2, \dots, S_n \in \mathcal{F}$ (i.e. $S_i \cap S_j = \emptyset$ for $i \neq j$) and real numbers $a_1, a_2, \dots, a_n$ such that

$$f(x) = \sum_{k=1}^n a_k \mathbb{1}_{S_k}(x) \quad \text{for any } x \in \Omega.$$

Now let $f : \Omega \rightarrow \mathbb{R}$ be a simple function having values a_k on the sets S_k , denote

$$I(f) = \sum_{k=1}^n a_k \mu(S_k).$$

Note that $I(f)$ does not depend on the partition $\{S_k\}_{k=1}^{\infty}$ (why?), so $I(\cdot)$ is a **well-defined** function on the **set of all simple functions**.

Similarly for any set $E \in \mathcal{F}$ we denote

$$I_E(f) = \sum_{k=1}^n a_k \mu(S_k \cap E),$$

and it can be shown that $I_E(\cdot)$ is also well-defined in the case of any $E \in \mathcal{F}$.

Besides the measure space $(\Omega, \mathcal{F}, \mu)$ consider the measure space $(\mathbb{R}, \mathcal{M}, \lambda)$, where $\mathcal{M}$ and λ are the Lebesgue algebra and Lebesgue measure respectively. Now let $f : (\Omega, \mathcal{F}, \mu) \rightarrow (\mathbb{R}, \mathcal{M}, \lambda)$ be a measurable function from Ω to $\mathbb{R}$, i.e. $f^{-1}(B) \in \mathcal{F}$ for any measurable set $B \in \mathcal{M}$, and assume $f(x) \geq 0$ for any $x \in \Omega$.

Consider the quantities

$$I_*(f) = \sup\{I(g) \mid g \text{ is a simple function such that } 0 \leq g(x) \leq f(x) \text{ for any } x \in \Omega\}$$

and

$$I_{*E}(f) = \sup\{I_E(g) \mid g \text{ is a simple function such that } 0 \leq g(x) \leq f(x) \text{ for any } x \in \Omega\}.$$

Exercise 3. Show that for a simple function f we will have $I_*(f) = I(f)$ and $I_{*E}(f) = I_E(f)$.

Definition 9 (Lebesgue integral for non-negative functions). If $f : \Omega \rightarrow \mathbb{R}$ is a non-negative measurable function and $E \in \mathcal{F}$ then $I_*(f)$ and $I_{*E}(f)$ are called the **Lebesgue integrals** of f on Ω and E respectively and are denoted by

$$I_*(f) = \int_{\Omega} f d\mu, \quad I_{*E}(f) = \int_E f d\mu.$$

Now let $f : \Omega \rightarrow \mathbb{R}$ be arbitrary measurable function (i.e. not necessarily non-negative), then we define

$$f^+(x) = \begin{cases} f(x) & \text{if } f(x) \geq 0 \\ 0 & \text{otherwise,} \end{cases} \quad f^-(x) = \begin{cases} -f(x) & \text{if } f(x) \leq 0 \\ 0 & \text{otherwise,} \end{cases}$$

Definition 10 (Lebesgue integral). We say the measurable function f is **Lebesgue integrable** if the following integrals are finite:

$$\int_{\Omega} f^+ d\mu < \infty, \quad \int_{\Omega} f^- d\mu < \infty.$$

In that case the **Lebesgue integral** on Ω is defined by

$$\int_{\Omega} f d\mu = \int_{\Omega} f^+ d\mu - \int_{\Omega} f^- d\mu.$$

Similarly, the following quantity is called **Lebesgue integral** on the measurable set $E \in \mathcal{F}$:

$$\int_E f d\mu = \int_E f^+ d\mu - \int_E f^- d\mu.$$

The following notion of “almost everywhere” will be used in the further reading quite often.

Definition 11 (“almost everywhere”). We say something happens “almost everywhere” in a measure space $(\Omega, \mathcal{F}, \mu)$ if the set of $x \in \Omega$ for which the thing does not happen has 0 measure. For example we say $f : \Omega \rightarrow \mathbb{R}$ is non-negative **almost everywhere** if

$$\mu(\{x \in \Omega \mid f(x) < 0\}) = 0.$$

Exercise 4. Prove the following properties of the Lebesgue integral.

- **Absolute Integrability:** A function f is Lebesgue integrable if and only if $|f|$ is Lebesgue integrable:

$$f \text{ is integrable} \iff |f| \text{ is integrable.}$$

- **Linearity:** If f and g are Lebesgue integrable and $a, b \in \mathbb{R}$, then:

$$\int_E (af + bg) d\mu = a \int_E f d\mu + b \int_E g d\mu.$$

- **Monotonicity:** If $f \leq g$ almost everywhere on E and both are integrable, then:

$$\int_E f d\mu \leq \int_E g d\mu.$$

- **Non-Negativity:** If $f \geq 0$ almost everywhere on E , then:

$$\int_E f d\mu \geq 0.$$

- **Additivity Over Domains:** If A and B are disjoint (i.e. $A \cap B = \emptyset$) measurable subsets of E , then:

$$\int_{A \cup B} f d\mu = \int_A f d\mu + \int_B f d\mu.$$

- **Dominated Convergence Theorem:** If $f_n \rightarrow f$ almost everywhere on E and $|f_n| \leq g$, where g is integrable, then:

$$\lim_{n \rightarrow \infty} \int_E f_n d\mu = \int_E f d\mu.$$

1.3 Probability and random variables

Definition 12 (probability space). Let $(\Omega, \mathcal{F})$ be a measurable space, $\mathbb{P} : \mathcal{F} \rightarrow \mathbb{R}$ be a measure such that $\mathbb{P}(\Omega) = 1$. Then $\mathbb{P}$ is called a **probability measure** and $(\Omega, \mathcal{F}, \mathbb{P})$ a **probability space**. Also for the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ any set $B \in \mathcal{F}$ is called an **event**.

Definition 13 (conditional probability). Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $A, B \in \mathcal{F}$ be such events that $\mathbb{P}(B) > 0$. Then

$$\mathbb{P}(A|B) := \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

is called **the conditional probability of the event A given the event B (observed)**.

Definition 14 (random variable (vector)). Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, $X : \Omega \rightarrow \mathbb{R}^n$ be a measurable function (w.r.t. the measure space $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n), \lambda)$, where $\mathcal{B}(\mathbb{R}^n)$ is the Borel σ -algebra and λ is the Lebesgue measure), i.e. $X^{-1}(B) \in \mathcal{F}$ for any $B \in \mathcal{B}(\mathbb{R}^n)$. Then X is called a **random variable**.

Definition 15 (discrete random variable). Let $X : \Omega \rightarrow \mathbb{R}^n$ be a random variable with image being a **finite or countable set**, i.e. there are such vectors $x^1, \dots, x^m, \dots \in \mathbb{R}^n$ such that $X(\omega) \in \{x_m\}_{m=1}^\infty$ for any $\omega \in \Omega$. Then we say X is a **discrete random variable**.

Definition 16 (probability mass function). Let $X : \Omega \rightarrow \mathbb{R}^n$ be a discrete random variable defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Then the following function is called the **(joint) probability mass function (pmf)**:

$$p_X : \mathbb{R}^n \rightarrow [0, 1], \quad p_X(x) = \mathbb{P}(\omega \in \Omega \mid X(\omega) = x).$$

Definition 17. Let $X : \Omega \rightarrow \mathbb{R}^n$ be a random variable, then the following σ -algebra is called **the σ -algebra generated by X** :

$$\sigma(X) = \{X^{-1}(B) \mid B \in \mathcal{B}(\mathbb{R}^n)\},$$

and **this is the minimal σ -algebra w.r.t. which X is measurable**.

The following lemma holds:

Lemma 1 (Doob-Dynkin). Let $X, Y : \Omega \rightarrow \mathbb{R}^n$ be random variables. Then Y is $\sigma(X)$ -measurable (i.e. $Y^{-1}(B) \in \sigma(X)$ for any $B \in \mathcal{B}(\mathbb{R}^n)$) **if and only if** there is a Borel measurable function $g : \mathbb{R}^n \rightarrow \mathbb{R}^n$ (i.e. $g^{-1}(B) \in \mathcal{B}(\mathbb{R}^n)$ for any $B \in \mathcal{B}(\mathbb{R}^n)$) such that $Y = g(X)$.

Definition 18. Let $X : \Omega \rightarrow \mathbb{R}^n$ be a random variable in the probability space $(\Omega, \mathcal{F}, \mathbb{P})$, then the following function

$$P_X : \mathcal{B}(\mathbb{R}^n) \rightarrow \mathbb{R}, \quad B \mapsto \mathbb{P}(X^{-1}(B))$$

will be a **probability** on the measurable space $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$ (show why) and is called the **distribution of X** .

Definition 19 (expectation). Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, $X : \Omega \rightarrow \mathbb{R}$ be a **Lebesgue integrable random variable**, then the following integral is called **the expectation of X** :

$$\mathbb{E}[X] = \int_{\Omega} X d\mathbb{P}.$$

It can be shown that if X is a random variable with $\mathbb{E}[X] < \infty$ then, since P_X is a measure on $\mathbb{R}$, the following equality holds:

$$\mathbb{E}[X] = \int_{\mathbb{R}} x dP_X.$$

Definition 20 (absolutely continuous random variable, density function). A random variable $X : \Omega \rightarrow \mathbb{R}$ is called **absolutely continuous** if there is a function $p_X : \mathbb{R} \rightarrow \mathbb{R}$ such that for any real numbers $a < b$

$$\mathbb{P}(\{\omega \in \Omega \mid X(\omega) \in [a, b]\}) = \int_{[a, b]} p_X d\lambda = \int_a^b p_X(x) dx,$$

where λ is the Lebesgue measure, so $\int_{[a, b]} p_X d\lambda$ is the Lebesgue integral of the function p_X , and $\int_a^b p_X(x) dx$ is the Riemann integral of p_X on the interval $[a, b]$. It is well known that when p_X is bounded and is integrable by Riemann on an interval $[a, b]$ then it is also Lebesgue integrable on $[a, b]$ and the Lebesgue integral is equal to the Riemann integral: $\int_{[a, b]} p_X d\lambda = \int_a^b p_X(x) dx$.

The function $p_X(\cdot)$ is called the **probability density function (pdf)** of the random variable X .

Definition 21 (density in higher dimensions). Let $X : \Omega \rightarrow \mathbb{R}^n$ be a (n -dimensional) random variable. If there is a function $p_X : \mathbb{R}^n \rightarrow \mathbb{R}$ such that for any Borel set $A \in \mathcal{B}(\mathbb{R}^n)$

$$\mathbb{P}(\{\omega \in \Omega \mid X(\omega) \in A\}) = \int_A p_X d\lambda,$$

where λ is the Lebesgue measure on $\mathbb{R}^n$, then the function p_X is called the **density function of the random variable X** .

The density of X is also called the **joint density of $X_1, X_2, \dots, X_n$** , where $X_i : \Omega \rightarrow \mathbb{R}$ are the projections of X , i.e. for any $\omega \in \Omega$

$$X(\omega) = \begin{bmatrix} X_1(\omega) \\ X_2(\omega) \\ \vdots \\ X_n(\omega) \end{bmatrix}.$$

EXAMPLE (multivariate Gaussian density): Let $X : \Omega \rightarrow \mathbb{R}^n$ be an n -dimensional random variable. We say that X follows a **multivariate Gaussian (normal) distribution** with mean vector $\mu \in \mathbb{R}^n$ and covariance matrix $\Sigma \in \mathbb{R}^{n \times n}$ (where Σ is symmetric and positive definite) if its density function is given by:

$$p_X(x) = \mathcal{N}(x; \mu, \Sigma) = \frac{1}{\sqrt{(2\pi)^n |\Sigma|}} e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)},$$

where $|\Sigma|$ denotes the determinant of Σ . In the special case when $n = 1$, $\mu \in \mathbb{R}$, and $\Sigma = \sigma^2 > 0$, we get the univariate Gaussian density:

$$p_X(x) = \mathcal{N}(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$

When $\mu = 0$ and $\Sigma = I$ (the identity matrix), we call this the **standard Gaussian (normal) distribution** and denote it by $\mathcal{N}(0, I)$.

It can be shown that for any mean vector $\mu \in \mathbb{R}^n$ and any symmetric positive definite covariance matrix $\Sigma \in \mathbb{R}^{n \times n}$, there exists a random variable $X : \Omega \rightarrow \mathbb{R}^n$ with the density function $\mathcal{N}(x; \mu, \Sigma)$.

Connections between the concepts of Lebesgue integral and Riemann integral lead to the following notations also for the Lebesgue integral. If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a measurable and Lebesgue-integrable function then the Lebesgue integral on a Borel set $A \subseteq \mathbb{R}^n$ will also be denoted by

$$\int_A f d\lambda = \int_A f(x) dx = \int \int \cdots \int_A f(x_1, \dots, x_n) dx_1 dx_2 \dots dx_n$$

Now let $X : \Omega \rightarrow \mathbb{R}^n$ be a random variable, and $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a (Borelian) function, then

$$Y : \Omega \rightarrow \mathbb{R}, \quad Y(\omega) = f(X(\omega))$$

is also a random variable and is denoted by $Y = f(X)$. In that case the following relation holds which we will use quite often in future:

$$\mathbb{E}[Y] = \mathbb{E}[f(X)] = \int_{\Omega} f(X) d\mathbb{P} = \int_{\mathbb{R}^n} f(x) p_X(x) dx.$$

Now let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and $\mathcal{H}_1, \dots, \mathcal{H}_n \subseteq \mathcal{F}$ be arbitrary subsets of $\mathcal{F}$. Also let us denote $\mathcal{A} = \{\mathcal{H}_1, \dots, \mathcal{H}_n\}$.

Definition 22 (independence). A collection $\mathcal{A} = \{\mathcal{H}_i\}_{i=1}^n$ is called **independent** if for any indices $0 < i_1 < \dots < i_k \leq n$ and any sets $H_{i_1} \in \mathcal{H}_{i_1}, \dots, H_{i_k} \in \mathcal{H}_{i_k}$ the following equality holds:

$$\mathbb{P}(H_{i_1} \cap \dots \cap H_{i_k}) = \mathbb{P}(H_{i_1}) \cdot \dots \cdot \mathbb{P}(H_{i_k}).$$

When $\mathcal{H}_i$ is a set of one element from $\mathcal{F}$, i.e. $\mathcal{H}_i = \{H_i\}$ for any $i = 1, \dots, n$, and $\mathcal{A} = \{\mathcal{H}_i\}_{i=1}^n$ is independent, then it is said that **the events $H_1, \dots, H_n$ are independent**.

Definition 23 (independence of random variables). A collection (sequence) of random variables $X_1, X_2, \dots$ ($X_i : \Omega \rightarrow \mathbb{R}^n$) is said to be **independent** if the collection $\mathcal{A} = \{\sigma(X_i)\}_{i=1}^\infty$ is independent.

Lemma 2. If $X, Y : \Omega \rightarrow \mathbb{R}$ are independent then

- $p_{X,Y}(x, y) = p_X(x) \cdot p_Y(y)$;
- $\mathbb{E}[X \cdot Y] = \mathbb{E}[X] \cdot \mathbb{E}[Y]$,

where $p_{X,Y}(\cdot, \cdot)$ is the joint density of the random variables X and Y .

Definition 24 (conditional density). Let X, Y be random variables with densities $p_X : \mathbb{R}^m \rightarrow \mathbb{R}$ and $p_Y : \mathbb{R}^l \rightarrow \mathbb{R}$ respectively such that $p_Y(y) > 0$ for any $y \in \mathbb{R}^l$. Also let $p_{(X,Y)}(x, y)$ be the joint density of $(X, Y) = (X_1, \dots, X_m, Y_1, \dots, Y_l)$. Then the function

$$p_{X|Y} : \mathbb{R}^{m+l} \rightarrow \mathbb{R}, \quad p_{X|Y}(x|y) = \frac{p_{(X,Y)}(x, y)}{p_Y(y)}$$

is called **the conditional density of X given Y** .

Definition 25 (Markov Chain). Let $X_1, X_2, \dots$ be a sequence of random variables. Then we call the sequence **Markov chain** if

$$p(x_t | x_1, \dots, x_{t-1}) = p(x_t | x_{t-1}) \quad \text{for any } t = 2, 3, \dots,$$

where $p(x_t | x_1, \dots, x_{t-1})$ (for any choice of $t, t_i, n \in \mathbb{N}$) is the conditional density function of the random variable X_t given the random variables $X_{t_1}, \dots, X_{t_n}$.

Definition 26 (stochastic processes). Let $T > 0$ be a positive real number, and $\{X_t \mid t \in [0, T]\}$ be a set of random variables defined on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Then $\{X_t\}_{t \in [0, T]}$ is called a **stochastic process**.

Here t usually denotes the time, so at each time t we have a random variable X_t . This can be useful when investigating such processes in nature when at each point of time the state of a system can be random but should obey some rules (the distributions/densities of X_t). If we fix some element from the probability space $\omega \in \Omega$, and consider the random process at that point we will have the process gone through some fixed “path”, therefore the function

$$[0, T] \rightarrow \mathbb{R}, \quad t \mapsto X_t(\omega)$$

is called **a path of $\{X_t\}_{t \in [0, T]}$** .